ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

CONCEPT RECAPITULATION TEST - I

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 02 and 09 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (03 to 08) contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 marks** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

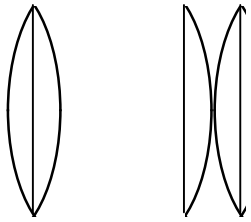
1. Number of amplitude modulation broadcast stations that can be accommodated in a 100 KHz bandwidth if the highest frequency modulating a carrier is 5 KHz
   (A) 10 (B) 100
   (C) 1000 (D) 10,000

2. Regarding transistor what is not correct.
   (A) For transistor to act an amplifier EB junction should be forward biased and CB junction should be reverse biased
   (B) $I_E = I_B + I_C$ in any configuration and for any transistor.
   (C) $\alpha = \frac{\beta}{1+\beta}$ where α and β transistor parameter
   (D) $\beta = \frac{\alpha}{1+\alpha}$.

3. A particle of charge q and mass m is projected from a large distance towards another identical charged particle at rest with velocity v_0. The distance of closest approach will be (Assume negligible gravitational attraction)
   (A) $\frac{q^2}{2\pi\varepsilon_0 m v_0^2}$ (B) $\frac{q^2}{\pi\varepsilon_0 m v_0^2}$
   (C) $\frac{2q^2}{\pi\varepsilon_0 m v_0^2}$ (D) $\frac{q^2}{4\pi\varepsilon_0 m v_0^2}$

Space for Rough work

4. A thin equiconvex lens of focal length f is cut into two symmetric plano-convex lenses, and the lenses are placed with their curved faces towards each other, with a common optic axis. If the refractive index μ of the material of the lens is 1.45, the focal length of the combination is (assuming no reflection at the boundaries) :

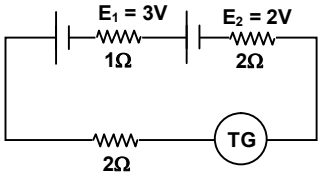


(A) f (B) $-f$
(C) 0.9 f (D) + f/0.9

5. In an npn transistor amplifier circuit when working in common emitter configuration when the input signal changes by 0.02 V, the base current changes by 10 μA and the collector 1 μA. If the load in the collector circuit is 10 KΩ for this the power given.
(A) 10^4 (B) 2×10^4
(C) 5×10^4 (D) 5×10^3

6. Two nicols A and B are placed in the path of a beam of unpolarised light. In between these two a third nicol C is placed such that its principal section is at an angle of 30° with that of A. The percentage of intensity of incident unpolarized light that emerges from C to B.
(A) 2.8% (B) 9.4%
(C) 15.3% (D) 10.2%

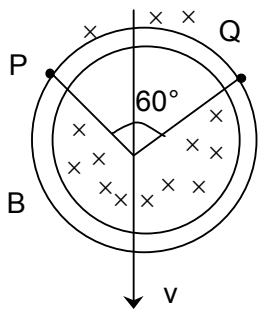
7. Two cells are connected with 2Ω resistance and a TG(tangent galvanometer). If the deflection in TG is 30°. Then the resistance of the coil.
Given that coil has 5 turns and radius = 0.08 and
$B_H = 5 \times 10^{-5}$T



(A) 1.5 Ω (B) 3.5 Ω
(C) 5.5 Ω (D) 7.5 Ω

Space for Rough work

8. A conducting ring of radius a falls vertically downward with a velocity v in a magnetic field B. The potential difference between two points P and Q located symmetrically on both sides of the vertical has the value



(A) 0 (B) $\frac{Bv_0a}{2}$

(C) $\frac{\sqrt{3}}{2}Bv_0a$ (D) Bv_0a

9. In problem 8, the current flowing in the upper arc PQ of the loop is given by (resistance per unit length is λ)

(A) 0 (B) $\frac{3Bv_0a}{2\pi\lambda}$

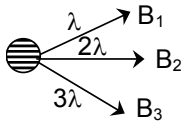
(C) $\frac{3\sqrt{3}}{2\pi\lambda}Bv_0a$ (D) $\frac{3}{\pi\lambda}Bv_0a$

10. A parallel beam of light of all wavelengths greater than 3000 A^0 falls on a double slit in a Young's double slit experiment. It is observed that the wavelengths 3600 A^0 and 6000 A^0 are absent at a distance of 31.5 mm from the position of the central maximum, and the orders of the interference at this point for the two wavelengths differ by 7. If the distance between the slits and the screen is 1m, the separation between the two slits is

(A) 0.08 mm (B) 0.13 mm

(C) 0.2mm (D) 0.1mm

11. A certain radioactive material can undergo three different types of decay, each with a different decay constant λ, 2λ and 3λ. Then, the effective decay constant λ_{eff} is given by



(A) $\frac{1}{\lambda_{eff}} = \frac{1}{\lambda} + \frac{1}{2\lambda} + \frac{1}{3\lambda}$ (B) $\lambda_{eff} = \lambda + 2\lambda + 3\lambda$

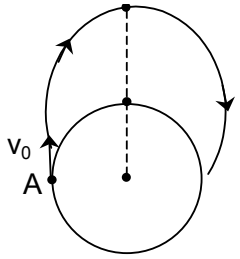
(C) $\lambda_{eff} = \frac{\lambda + 2\lambda + 3\lambda}{3}$ (D) $\frac{1}{\lambda_{eff}} = \frac{1}{3}\left(\frac{1}{\lambda} + \frac{1}{2\lambda} + \frac{1}{3\lambda}\right)$

Space for Rough work

12. A thin walled cylindrical metal vessel ($\alpha = 10^{-3}K^{-1}$) contains a liquid ($\gamma_r = 10^{-3}K^{-1}$). If the vessel and its contents are now heated by 10 K the pressure due to the liquid at the bottom

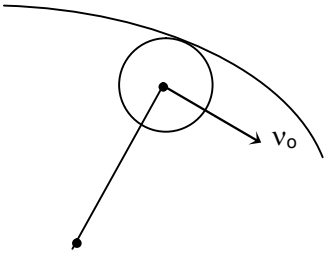
(A) increases by 2 %
(B) decreases by 1 %
(C) remains unchanged
(D) decreases by 2%

13. A projectile projected tangentially from the surface of a planet of radius R from a point A, follows the indicated trajectory during its motion. If it is at a height of 3R at the farthest point of its trajectory, then the velocity of projection at A, v_0 is given by (acceleration due to gravity on surface = g)



(A) $v_0 = \sqrt{1.5gR}$
(B) $v_0 = \sqrt{0.5gR}$
(C) $v_0 = \sqrt{1.6gR}$
(D) $v_0 = \sqrt{2gR/3}$

14. A small uniform sphere of diameter d is rolling without slipping with in another static hollow sphere of inner diameter 6d, the velocity of the centre of mass of the inner sphere being v_0 in magnitude. The angular velocity of the smaller sphere about an axis passing through its C.M. perpendicular to its direction of motion is



(A) $\frac{2v_0}{5d}$
(B) $\frac{2v_0}{d}$
(C) $\frac{12v_0}{5d}$
(D) $\frac{14v_0}{5d}$

15. 1 mole of H_2 and 1 mole of He at a pressure of P_0 & $2P_0$ respectively and a temperature T_0 are kept in two different containers. The ratio of their rms velocities is $C_{H_2} : C_{He}$ =

(A) $\sqrt{10} : \sqrt{3}$
(B) 1 : 1
(C) 5 : 3
(D) 2 : $\sqrt{2}$

Space for Rough work

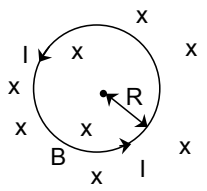
16. The wavelength of the K_α line for an element is measured to be λ = 1.55 A^0. The element has Z (atomic number) equal to
(A) 28 (B) 27
(C) 29 (D) 30

17. An equimolar mixture of a monoatomic and a diatomic ideal gas is suddenly compressed to $\frac{1}{8}^{th}$ of its original volume. The ratio of final temperature to the initial temperature.
(A) $8^{0.53}$ (B) $8^{0.5}$
(C) $8^{1.53}$ (D) $8^{2/3}$

18. The material of an electromagnet should have following properties, except
(A) high retentivity
(B) high value of saturation magnetisation
(C) low coercivity
(D) low hystersis loss

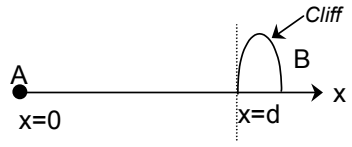
19. According to electromagnetic theory statement that is not correct.
(A) $\vec{E}$ and $\vec{B}$ vectors vibrate sinusoidally in the same phase
(B) $\vec{E}$ and $\vec{B}$ vectors vibrate sinusoidally in the opposite phase
(C) $\vec{E}$ and $\vec{B}$ vectors contribute to energy equally
(D) $\vec{E}$ and $\vec{B}$ vectors vibrate orthogonally with the same frequencies

20. A current (I) carrying circular wire of radius R is placed in a magnetic field B perpendicular to its plane. The change in tension T of the wire is
(A) BIR (B) 2πBIR
(C) πBIR (D) 2BIR



Space for Rough work

21. The velocity of sound varies according to the relation :
$c(x) = c_0 (1 + 0.1x/d)$ between a source located at $x = 0$ and a cliff located at $x = d$. The time taken for sound to travel from A to B and return back to A is given by



(A) $\frac{2d}{c_0}$ (B) $\frac{2d}{c_0(1+\frac{0.1}{2})}$

(C) $\frac{2d}{c_0 \times 1.1}$ (D) $\frac{20d}{c_0}\ln(1.1)$

22. A charged particle of mass m and charge e moves in a circular path of radius r under the action of a constant magnetic field B acting perpendicular to the plane of its orbit. If Bohr's postulate is valid for this system, then the radius of the path, r is given by
(A) $r \propto n^2$ (B) $r \propto 1/n$
(C) $r \propto e^{-n}$ (D) $r \propto \sqrt{n}$
where n is the quantum number of the orbit.

23. The molar heat capacity of a diatomic ideal gas undergoing the process :
$PV^{1/4}$ = constant is
(A) $\frac{17}{6}R$ (B) $\frac{2}{5}R$
(C) $\frac{23}{6}R$ (D) 0

24. A wheel of radius r rolls without slipping on a flat surface, with its centre of mass undergoing a linear deceleration a. Then,
(A) a non -zero toque is acting on the wheel about its CM.
(B) the net torque acting on the wheel may be zero or non-zero
(C) the net torque acting on the wheel is zero
(D) the net torque acting on the wheel cannot be calculated as the data is insufficient.

Space for Rough work

25. A capacitor C and a resistance R are connected in a circuit in series with each other, so that the capacitor discharges through R. If the temperature coefficient of the resistance is positive, as the temperature increases the time constant of the circuit
(A) increases
(B) decreases
(C) remains uncharged
(D) may increase or decrease depending on other factors

26. A particle A having charge q is placed at a distance d from another particle B having an equal charge q an dielectric slab of thickness t is placed midway between the two. Electrostatic force on A due to B
(A) decreases
(B) increases
(C) remains constant
(D) may increase or decrease depending upon the magnitudes of the charges

27. A radioactive element A decays into another element B with a decay constant 4λ, which then decays into a third element C (decay constant = 9λ). If all the atoms in the beginning consisted of atoms of type A only, then the ratio of the number of atoms of A to that of type B (i.e. N_A/N_B) when the number of atoms of B is at a maximum (N_B is maximum) is
(A) $\frac{3}{2}$ (B) $\frac{9}{4}$
(C) $\frac{1}{1}$ (D) $\frac{\ln 3}{\ln 2}$

28. If planck's constant (h) were to be doubled while other fundamental quantities like and charge mass electron. remaining constant the force acting on the electron in the nth orbit of a hydrogen atom will
(A) increases 4 times (B) decrease 1/8 of its value
(C) remains unchanged (D) decreases 1/16 of its original value

29. A positron -electron pair form a hydrogen like system. Applying Bohr's theory to this system and ignoring any relativistic effects, the distance (a) between the positron and electron in the ground state is given by
(A) $a > 0.53\ A^0$ (B) $a < 0.53\ A^0$
(C) $a = 0.53\ A^0$ (D) $a = 1.06\ A^0$

30. Two identical spherical stars each having a mass equal to that of the sun move in a common circular orbit of radius equal to the earth's orbital radius, under their mutual gravitational interaction. Their time period of rotation equals
(A) 2yr (B) 1 yr
(C) $2\sqrt{2}$ yr (D) $\sqrt{2}$ yr

Space for Rough work

Chemistry

PART – II

SECTION – A

Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A gas in an open container is heated from 27°C to 127°C. The fraction of the original amount of remaining in the container will be:
   (A) $\frac{3}{4}$ (B) $\frac{1}{4}$
   (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

2. Which one has peroxide linkage
   (A) $S_2O_8^{2-}$ (B) $S_2O_6^{2-}$
   (C) $S_2O_7^{2-}$ (D) $S_2O_3^{2-}$

3. The number of electrons gained in the following reaction $Fe_2(SO_4)_3 \longrightarrow Fe^{2+} + SO_2$ is
   (A) 2 (B) 1
   (C) 6 (D) 8

4. The values of observed and calculated molecular weights of silver nitrate are 92.64 and 170 respectively. The degree of dissociation of silver nitrate is
   (A) 60% (B) 83.5%
   (C) 46.7% (D) 60.23%

5. The pH of pure water at 25°C and 35°C are 7 and 6 respectively. The heat of formation of water from H^+ and OH^- are
   (A) 84.55 kcal mol^{-1} (B) – 84.55 kcal $mole^{-1}$
   (C) 74.55 kcal mol^{-1} (D) – 74.55 kcal $mole^{-1}$

6. In a face centred cubic arrangement of A and B atoms whose A atoms are at the corner of the unit cell and B atoms are at the face centres. One of the A atom is missing from one corner in unit cell. The simplest formula of the compound is:
   (A) A_7B_3 (B) AB_3
   (C) A_7B_{24} (D) $A_{7/8}B_3$

Space for Rough work

7. In which one of the following reactions, we will get only one crossed Aldol product,
(A) C_6H_5CHO, $(CH_3)_2CO$ (B) C_6H_5CHO, CH_3CHO
(C) C_6H_5CHO, $(C_2H_5)_2CO$ (D) CH_3CHO, CH_3CH_2CHO

8. $CH_3 - \underset{\underset{OH}{|}}{CH} - \underset{\underset{O}{||}}{C} - CH_3$. This compound
(A) Can react with Schiff's Reagent (B) Can react with Tollen's Reagent
(C) Both (A) and (B) (D) None

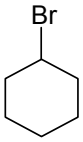
9. The isomeric monosubstitution products theoretically possible for the structure $CH_2 = CH - CH_2 - CH_2 - CH = CH_2$ are
(A) 3 (B) 2
(C) 4 (D) 6

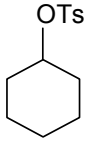
10. $CH_3 - \underset{\underset{O}{\diagdown\diagup}}{\overset{CH_3}{\overset{|}{C}}} - CH_2 \xrightarrow{CH_3OH / OH^-}$ A. A is

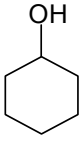
(A) $CH_3 - \overset{CH_3}{\overset{|}{C}}(OCH_3)(OH) - CH_2$ (B) $CH_3OH + CH_3 - \overset{CH_3}{\overset{|}{C}} = CH_2$

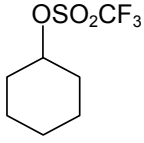
(C) $CH_3 - \overset{CH_3}{\overset{|}{C}}(OH)(OCH_3) - CH_2$ (D) $HCHO + CH_3 - \overset{CH_3}{\overset{|}{C}} = CH_2$

11. $HO - CH_2 - CH_2 - X + HC \equiv C^-Na^+ \longrightarrow A + B$
A & B can be
(A) $HO - CH_2 - CH_2 - C \equiv CH$, NaX (B) Acetylene + $NaO - CH_2 - CH_2 - X$
(C) Ethanol + $XC \equiv CNa^+$ (D) Ethylhalide + $HOC \equiv CNa^+$

12. Which of the following will be most reactive towards the nucleophilic substitution.
(A) 

(B) 

(C) 

(D) 

Space for Rough work

13. The correct order of hybridisation in the following species SO_2, ClO_3^-, XeF_4 is
(A) sp^3, sp^2, sp^3d (B) sp^2, sp^2, sp^3d^2
(C) sp^2, sp^3, sp^3d^2 (D) sp, sp^3, sp^3d^2

14. The species with maximum magnetic moment is
(A) $K_4[Ni(CN)_6]$ (B) $K_4[Fe(CN)_6]$
(C) $K_3[Co(CN)_6]$ (D) $[Cu(NH_3)_4]SO_4$

15. The increasing order of dipole moment is
(A) $CH_3I < CH_3Br < CH_3Cl < CH_3F$ (B) $CH_3F < CH_3Cl < CH_3Br < CH_3I$
(C) $CH_3I < CH_3Br < CH_3F < CH_3Cl$ (D) $CH_3I < CH_3F < CH_3Br < CH_3Cl$

16. 17g of hydrogen peroxide is present in 1120 ml of solution. This solution is called
(A) 5 volume solution (B) 10 volume solution
(C) 15 volume solution (D) 16 volume solution

17. The density of a gas at 27°C and 1 atm is d. Pressure remaining constant, at which of the following temperature will its density become 0.75d.
(A) 20°C (B) 30°C
(C) 36°C (D) 127°C

18. Radioactivity of radioactive element remains 1/10 of the original radioactivity after 2.303 seconds. The half life period (in seconds) is
(A) 2.303 (B) 0.693
(C) 0.2303 (D) 0.0693

19. Assuming complete dissociation predict which of the following will have pH = 12.
(A) 500 ml 0.005 M NaOH (B) 1000 ml 0.005 M $Ca(OH)_2$
(C) 100 ml 0.05 M $Ca(OH)_2$ (D) 50 ml 0.05 M NaOH

20. The percentage of metal in its oxide is 60. The percentage of bromine in metal bromide (oxidation state of metal remains the same) will be
(A) ~ 87 (B) ~ 97
(C) ~ 77 (D) ~ 80

Space for Rough work

21. 50g of $CaCO_3$ when treated with 60 ml of 20 N HCl at 300 K and 4 atm produces
(A) 22 g CO_2 (B) 44g CO_2
(C) 11g CO_2 (D) 88 g CO_2

22. A certain weak acid has a dissociation constant of 10^{-5}. The equilibrium constant for its reaction with a strong base is
(A) 10^{-9} (B) 10^{+14}
(C) 10^{9} (D) 10^{-5}

23. The set representing the correct order of 1st ionisation potential is
(A) Cu > Ag > Au (B) Au >Ag > Cu
(C) Ag > Cu > Au (D) Au > Cu > Ag

24. Acetophenone on reaction with p -nitroperbenzoic acid gives
(A) phenyl propionate (B) phenyl acetate
(C) methyl benzoate (D) benzophenone

25. The number of monochloro derivaties of 2-methoxy propane are possible are
(A) 2 (B) 1
(C) 3 (D) 4

26. The increasing order of bond angle in the following species is
(A) $F_2O < Cl_2O < H_2O$ (B) $Cl_2O < H_2O < F_2O$
(C) $F_2O < H_2O < Cl_2O$ (D) $H_2O < F_2O < Cl_2O$

27. 0.10g of a sample containing $CuCO_3$ and some inert impurity was dissolved in dilute sulphuric acid and volume made up to 50 ml. This solution was added into 50 ml of 0.04 M KI solution where copper precipitates as CuI and I^- is oxidised into I_3^-. A 10 ml portion of this solution is taken for analysis filtered and made up free I_3^- and then treated with excess of acidic permanganate solution. Liberated iodine required 20 ml of 2.5 mM sodium thiosulphate solution to reach the end point. Determine the wt. percentage of $CuCo_3$ in the original sample.
(A) 7.41 (B) 74.1
(C) 61.75 (D) None of these

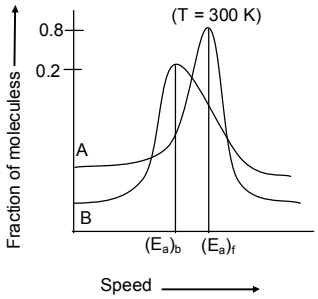
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28. Urea (H_2NCONH_2) is manufactured by passing $CO_2(g)$ through ammonia solution followed by crystallization. CO_2 for the above reaction is prepared by combustion of hydrocarbon. If combustion of 236 kg of a saturated hydrocarbon (C_nH_{2n+2}) produces as much CO_2 as required for production of 999.6 kg urea then M. F. of hydrocarbon is:
(A) $C_{10}H_{22}$ (B) $C_{12}H_{26}$
(C) $C_{13}H_{28}$ (D) C_8H_{18}

29. For the reaction at 300 K

$A \rightleftharpoons B \qquad \Delta H = -2400 \text{ cal mol}^{-1}$

Frequency factor for both forward and backward reaction is equal. The $[K_{eq}]_{400\ K}$ is $4e^{-0.5n.}$
The value of 'n' is (Assume ΔH independent of temperature)



(A) 1 (B) 2
(C) 4 (D) 6

30. Which of the following statement is not true?
(A) O-F bond length in OF_2 is less than O-F bond length in O_2F_2.
(B) In HCO_3^-, all C – O bond length are not identical.
(C) In diborane, two different B – H bond lengths are observed although the hybridization of both boron atoms are same.
(D) In hydrazine, the N – N bond length is larger than normal N – N bond length.

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Mathematics

PART – III

SECTION – A
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Let y = f(x) be a curve passing through (e, e^e), which satisfy the differential equation $(2ny + xy \ln x)\,dx - x \ln x\,dy = 0$, $x > 0$, $y > 0$. It $g(x) = \lim_{n\to\infty} f(x)$ then $\int_{1/e}^{e} g(x)dx$ equals to

   (A) e (B) 1
   (C) 0 (D) none of these

2. The smaller area enclosed by y = f(x), when f(x) is a polynomial of least degree satisfying $\lim_{x\to 0}\left(1+\frac{f(x)}{x^3}\right)^{1/x} = e$ and the circle $x^2 + y^2 = 2$ above the x-axis is

   (A) $\frac{\pi}{2}$ (B) $\frac{3}{5}$
   (C) $\frac{\pi}{2}-\frac{3}{5}$ (D) $\frac{\pi}{2}+\frac{3}{5}$

3. For $x \in R$ and a continuous function f, let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x f\{x(2-x)\}dx$,

   $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f\{x(2-x)\}dx$. Then $\frac{I_1}{I_2}$ is

   (A) 0 (B) 1
   (C) 2 (D) 3

Space for rough work

4. If $xf(x) = 3(f(x))^2 + 2$, then $\int \frac{2x^2 - 12xf(x) + f(x)}{(6f(x) - x)(x^2 - f(x))^2} dx$ equals to

(A) $\frac{1}{x^2 - f(x)} + c$ (B) $\frac{1}{x^2 + f(x)} + c$

(C) $\frac{1}{x - f(x)} + c$ (D) $\frac{1}{x + f(x)} + c$

5. If $f\left(2x + \frac{y}{8}, 2x - \frac{y}{8}\right) = xy$, then $f(m, n) + f(n, m) = 0$

(A) only when $m = n$ (B) only when $m \neq n$
(C) only when $m = -n$ (D) for all m and n

6. If period of $\sin^{2m}\sqrt{k}x$, $m \in N$ is π then $\lim_{n\to\infty} k^n$ is equal to

(A) 0 (B) ∞
(C) 1 (D) none of these

7. A function g(x) is defined as $g(x) = \frac{1}{4}f(2x^2 - 1) + \frac{1}{2}f(1 - x^2)$ and f′(x) is an increasing function then g(x) is increasing in the interval

(A) $\left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$ (B) $\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\frac{\sqrt{2}}{3}, \infty\right)$

(C) (–1, 1) (D) none of these

8. If $\sin\theta$ and $-\cos\theta$ are the roots of the equation $ax^2 - bx - c = 0$, where a, b, c are the sides of a triangle ABC, then cos B is equal to

(A) $1 - \frac{c}{2a}$ (B) $1 - \frac{c}{a}$

(C) $1 + \frac{c}{2a}$ (D) $1 + \frac{c}{3a}$

Space for rough work

9. The equation $\int_0^x \left(t^2 - 8t + 13\right)dt = x\sin\left(\frac{a}{x}\right)$ has a solution if $\sin\frac{a}{6}$ is equal to

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) 0

10. If $\vec{x}$ and $\vec{y}$ are two non-collinear vectors and a triangle ABC with side lengths a, b, c satisfying
$(20a - 15b)\vec{x} + (15b - 12c)\vec{y} + (12c - 20a)(\vec{x} \times \vec{y}) = \vec{0}$. Then triangle ABC is

(A) an acute angle triangle (B) an obtuse angle triangle
(C) a right angle triangle (D) an isosceles triangle

11. The Cartesian equation of the plane $\vec{r} = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$ is

(A) 2x + y = 5 (B) 2x – y = 5
(C) 2x + z = 5 (D) 2x – z = 5

12. If the ellipse $\frac{x^2}{4} + y^2 = 1$ meets the ellipse $x^2 + \frac{y^2}{a} = 1$ in four distinct points and $a = b^2 - 5b + 7$, then b does not lie in the interval

(A) [4, 5] (B) $(-\infty, 2) \cup (3, \infty)$
(C) $(-\infty, 0)$ (D) [2, 3]

13. Portion of asymptote of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (between centre and the tangent at vertex) in the first quadrant is cut by the line $y + \lambda(x - a) = 0$ (λ is a parameter) then

(A) $\lambda \in R$ (B) $\lambda \in (0, \infty)$
(C) $\lambda \in (-\infty, 0)$ (D) $\lambda \in R - \{0\}$

14. A focal chord of parabola $y^2 = 4x$ is inclined at an angle of $\frac{\pi}{4}$ with positive x-direction, then the slope of normal drawn at the ends of chord will satisfy the equation

(A) $m^2 - 2m - 1 = 0$ (B) $m^2 + 2m - 1 = 0$
(C) $m^2 - 1 = 0$ (D) $m^2 + 2m - 2 = 0$

Space for rough work

15. The range of values of a such that the angle θ between the pair of tangents drawn from (a, 0) to the circle $x^2 + y^2 = 1$ lies in the interval $\left(\frac{\pi}{3}, \pi\right)$ is

(A) $(-2, -1) \cup (1, 2)$ (B) $\left(-\sqrt{2}, 0\right) \cup \left(0, \sqrt{2}\right)$

(C) $\left(-\sqrt{3}, -\sqrt{2}\right)$ (D) $\left(-\sqrt{3}, -\sqrt{2}\right) \cup \left(\sqrt{2}, \sqrt{3}\right)$

16. The equation of the parabola to which the line $m^2(y - 10) - mx - 1 = 0$ is a tangent for any real value of m is

(A) $x^2 = -4y$ (B) $x^2 = -4(y - 10)$

(C) $y^2 = 4(x - 4)$ (D) $x^2 = y - 10$

17. Consider a curve $ax^2 + 2hxy + by^2 - 1 = 0$ and a point P not on the curve. A line drawn from the point P intersects the curve at the points Q and R. If the product PQ·PR is independent of the slope of the line, then the curve is

(A) a pair of straight lines (B) a circle

(C) a parabola (D) an ellipse or a hyperbola

18. In a ΔABC if A = (1, 2) and internal angle bisectors through B and C are y = x and y = –2x. The in radius r of ΔABC is

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{2}}$

(C) $\frac{2}{3}$ (D) none of these

19. If $0 \le [x] < 2$; $-1 \le [y] < 1$ and $1 \le [z] < 3$ (where [.] denotes the greatest integer function) then the maximum value of determinant $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is

(A) 2 (B) 6

(C) 4 (D) none of these

Space for rough work

20. If $\min\{x^2 + (a - b)x + 1 - a - b\} > \max\{-x^2 + (a + b)x - (1 + a + b)\}$, then
(A) $a^2 + b^2 < 2$ (B) $a^2 + b^2 < 4$
(C) $a^2 + b^2 > 4$ (D) $a + b > ab$

21. If m and x are two real numbers, then the real value of $e^{2mi\cot^{-1}x}\left(\frac{xi+1}{xi-1}\right)^m$ is equal to
(A) $\cos x + i \sin x$ (B) $\frac{m}{2}$
(C) 1 (D) $\frac{m}{2}+1$

22. If {x} represents the fractional part of x, then $\left\{\frac{5^{200}}{8}\right\}$ is
(A) $\frac{1}{4}$ (B) $\frac{1}{8}$
(C) $\frac{3}{8}$ (D) $\frac{5}{8}$

23. If $\frac{a_2a_3}{a_1a_4} = \frac{a_2+a_3}{a_1+a_4} = 3\left(\frac{a_2-a_3}{a_1-a_4}\right)$, then a_1, a_2, a_3, a_4 are in
(A) A.P. (B) G.P.
(C) H.P. (D) none of these

24. n-similar balls each of weight w when weighed in pairs the sum of the weights of all the possible pairs is 120 when they are weighed in triplets the sum of the weights comes out to be 480 for all possible triplets, then n is
(A) 20 (B) 10
(C) 16 (D) 28

Space for rough work

25. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\left|\vec{a}+2\vec{b}+3\vec{c}\right| = \sqrt{3+2\sqrt{2}}$. Angle between $\vec{a}$ and $\vec{b}$ is α, between $\vec{a}$ and $\vec{c}$ is β and angle between $\vec{b}$ and $\vec{c}$ varies in $\left[\frac{\pi}{2}, \frac{2\pi}{3}\right]$, then the greatest value of 4 cos α + 6 cos β is
(A) $2\sqrt{2}+5$ (B) $-2\sqrt{2}+5$
(C) $2\sqrt{2}-5$ (D) $2\sqrt{2}$

26. If two events A and B are such that $P(A^c) = 0.3$, $P(B) = 0.4$ and $P(A \cap B^c) = 0.5$ then $P\left(\frac{B}{A \cup B^C}\right)$ is equal to
(A) 0.9 (B) 0.5
(C) 0.6 (D) 0.25

27. ABC is a triangular park with AB = AC = 100m. A block tower is situated at the midpoint of BC. The angles of elevation of the top of the tower at A and B are $\cot^{-1}(3.2)$ and $\text{cosec}^{-1}(2.6)$ respectively. The height of the tower is
(A) 16 m (B) 25 m
(C) 50 m (D) none of these

28. If $aN = \{ax \mid x \in N\}$ and $bN \cap cN = dN$, where $b, c \in N$, then
(A) d = bc (B) c = bd
(C) b = cd (D) none of these

29. If the normal to the curve y = f(x) at x = 0, be given by the equation 3x – y + 3 = 0, then the value of $\lim_{x\to 0} x^2\left\{f\left(x^2\right)-5f\left(4x^2\right)+4f\left(7x^2\right)\right\}^{-1}$ is equal to
(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$
(C) $-\frac{1}{5}$ (D) $\frac{1}{4}$

30. Let P (n) be the statement $2^n < n!$ where n is a natural number, then P (n) is true for
(A) all n (B) all n > 2
(C) all n > 3 (D) none of these

Space for rough work